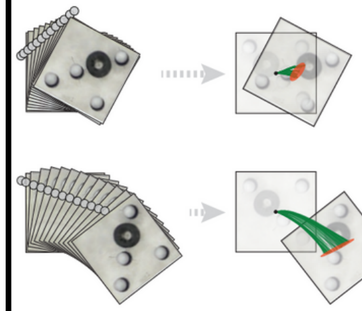
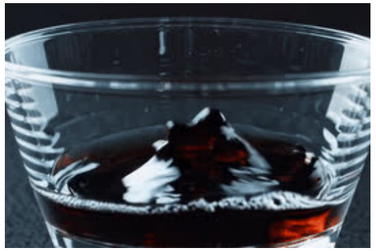
Modeling Burden

- It's hard to model dynamic phenomena!
  - Nonlinearities, stochasticity, etc.



Energetic buildup? Generalizability to other viscocities? →

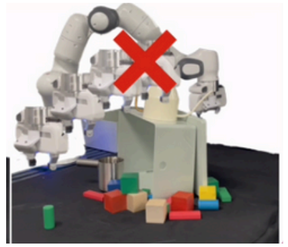
← Same push, distribution of outcomes; friction is stochastic



- Bad models → misguided planning
- Models for each physical phenomena?

Problem Dimensionality

- High-dimensional search (robot and object velocities, accelerations, forces) → slow
- Pre-computation in combined robot and task space → memory constraints
- Good first guess to solution? Best solution?

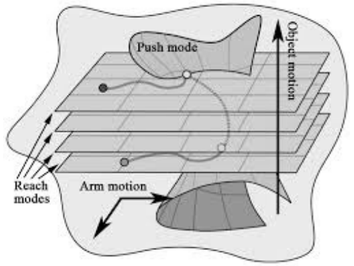
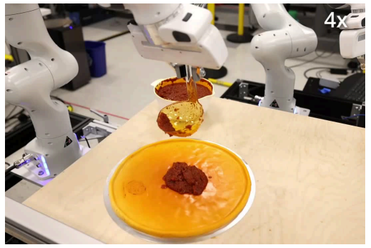


System Design

- Integration challenge with multiple specialized components (engineering complexity)
- Component tuning, error propagation, brittleness at system boundaries
- Lack of continual improvement

Usability

- Novel tasks → engineering hours
  - eg: task-specific sampling strategies for narrow passage problems



- How do you design a cost function? Right thing to model?